

Mahalingam R

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Skills

Languages: TypeScript, JavaScript, Python, C#, SQL, C/C++

Backend & Frameworks: NestJS, Node.js, FastAPI, ASP.NET Core MVC, Celery, Prisma ORM, SQLAlchemy

Databases & Caching: PostgreSQL, pgvector, Redis, Redis Streams, ElastiCache

Cloud & DevOps: AWS (Lambda, ECS Fargate, ECR, S3, SES, CloudFront, ALB, Route 53, ACM, Secrets Manager, EventBridge), GCP (Storage Transfer Service, Cloud Storage), Docker, GitHub Actions, CI/CD

AI Engineering: OpenAI GPT-4o, Gemini 2.0 Flash, Anthropic Claude, Whisper, Prompt Engineering, RAG, pgvector semantic search, BM25, Cross-Encoder Re-ranking, RAGAS, sentence-transformers, fastembed

Frontend Technologies: React, Next.js, TypeScript, Tailwind CSS, shadcn/ui, Material UI

Auth & Security: Auth0, JWT, RBAC, RSA-signed cookies, OAuth2

Integrations & Tools: Twilio (WhatsApp), Cashfree, Zoho Books, Stripe, SAP Integration Suite, PDFKit, Sharp, FFmpeg

Work Experience

Ekko Medical Solutions Pvt Ltd, Chennai

Oct 2025 - Present

Software Developer (Backend, AI & Distributed Systems)

- Architected real-time medical transcription pipeline using **WebSocket gateways (Soniox STT)**, **Redis Streams** with consumer groups for exactly-once processing, and background workers with configurable flush triggers (100-char / 15s / 5min timeout) for incremental persistence. Added batch audio processing with **FFmpeg** and speaker diarization.
- Designed **provider-agnostic LLM factory** with database-seeded primary/fallback configuration per tenant — supporting **OpenAI**, **Gemini**, and **Anthropic** models with automatic failover. Built **clinical extraction pipelines** transforming unstructured transcriptions into structured medical data with prompt engineering and JSON schema validation, persisted via **Prisma ORM** with transactional bulk upserts.
- Engineered **ophthalmology vitals extraction system** supporting images, PDFs, and audio — via **GPT-4o Vision** (images), **Gemini 2.0 Flash** (native PDF), and **Whisper** (audio-to-text → LLM); added Sharp compression, parallel processing, smart deduplication across 50+ categorized vital types, and event-driven non-blocking S3 uploads via **NestJS EventEmitter**.
- Implemented **Auth0-based JWT authentication** with granular **RBAC** (role + direct permission aggregation); replaced Auth0 default emails with custom **SES-branded invitations** backed by Management API password-change tickets (48-hour TTL) and **DKIM/SPF/DMARC** authentication.
- Developed **subscription billing system** integrating **Cashfree** (webhooks, reconciliation) with Luxon-based proration engine, 18% GST calculation, **Zoho Books API** for automated invoicing, and **Celery + Redis (ElastiCache TLS)** workers for license lifecycle management with at-least-once delivery guarantees.
- Implemented **CloudFront CDN** with **RSA-signed cookies** (24-hour expiry), **Origin Access Control**, and tenant-scoped S3 hierarchy for secure delivery of patient images, clinical documents, and audio — using **Secrets Manager** and **ACM wildcard SSL**.
- Delivered secure medical report distribution via **WhatsApp (Twilio)** and Email (**SES**) with PDFKit PDF generation, encrypted S3 storage, JWT-secured 24-hour expiring links, OTP fallback, and **DKIM/SPF/DMARC** authentication.
- Deployed on **ECS Fargate** behind **ALB** with target group routing, security groups (443/80), **Route 53 DNS**, **ACM certificates** — CI/CD via **GitHub Actions** pushing Docker images to **ECR** with zero-downtime rollouts. Automated **cross-cloud disaster recovery** using Lambda (pg_dump) → S3 (7-day retention) → GCP Storage Transfer Service to asia-south1 (30-day lifecycle) with **EventBridge** cron coordinated at 02:30 → 04:30 IST.

ROX Hi-TECH, Chennai

June 2024 - Oct 2025

Full Stack Developer

- Built SAP integrations using **SAP Integration Suite** — designed iFlows, managed APIs, and integrated **SAP S/4HANA** with internal applications (earned **SAP Certified Development Associate** certification).
- Developed full-stack HR data management application using **ASP.NET Core 8 (MVC)** with Razor views and **PostgreSQL** — implemented file upload forms, request-tracking workflow with auto-generated request numbers, and admin approve/reject functionality.
- Integrated SAP systems via **ODATA APIs** to fetch HR master data (payroll areas, personnel areas, gender data) through

ASP.NET Core backend services, surfacing the data into MVC views.

- Built admin dashboard module enabling master data management — infotype creation, role-based admin assignment, and end-to-end request approval/rejection workflow.
- Used **AJAX with jQuery** to send asynchronous POST requests from Razor views to MVC controller actions, enabling seamless form submissions without full page reloads.

Project Work

- **Full-Stack E-Commerce Platform (Shoppy)**

Live Demo

Built a production-grade e-commerce platform with Next.js 14 (App Router, Server Components, Server Actions) and NestJS backend deployed on AWS. Implemented JWT authentication with HttpOnly cookies and role-based access control (Admin/User), Stripe payment integration with webhook-driven order management, and product image uploads to AWS S3. Enabled real-time product updates via Socket.io WebSocket gateway. Set up CI/CD pipeline using AWS CodePipeline and CodeBuild for automated deployments to Elastic Beanstalk, with CloudFront CDN for edge caching and HTTPS termination. Designed a responsive UI with Tailwind CSS and Material UI.

Tech Stack: Next.js 14, NestJS, TypeScript, PostgreSQL, Prisma ORM, Stripe, AWS (Elastic Beanstalk, RDS, S3, CloudFront, CodePipeline, CodeBuild), Socket.io, Tailwind CSS, Material UI

- **MedRAG Assistant — Medical RAG Knowledge System**

Live Demo

Built a production-grade Retrieval-Augmented Generation (RAG) system for medical document Q&A using FastAPI and React. Implemented hybrid retrieval (BM25 + semantic search with pgvector) merged via Reciprocal Rank Fusion, cross-encoder re-ranking (ms-marco-MiniLM) to improve precision from top-20 candidates to top-5, and query rewriting for multi-turn conversations. Integrated Groq LLaMA 3.3 70B for grounded answer generation with source citations and SSE streaming. Built a RAGAS evaluation pipeline (OpenAI gpt-4o-mini judge) achieving 0.93 context recall and 0.86 faithfulness across 15 hand-written medical Q&A pairs. Deployed backend on Render, frontend on Vercel, PostgreSQL on Neon.

Tech Stack: FastAPI, Python, React, TypeScript, PostgreSQL, pgvector, fastembed (ONNX), Groq LLaMA 3.3 70B, RAGAS, Tailwind CSS, shadcn/ui

Education

SASTRA Deemed University

B.Tech. in Computer Science and Business Systems

Aug 2020 - Jun 2024

CGPA: 7.2/10

Certifications

- **SAP Certified Development Associate – SAP Integration Suite** — Validated skills in designing, developing, and managing integrations using SAP Integration Suite.